
How Wrong Can Your Simulator Be? Auditing, Tolerating, and Updating a Measured Forward Model on a Real Instrument

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Abstract

1 A learned restoration model for degraded macro X-ray fluorescence (MA-XRF)
2 scans of a painting is trained on a forward simulator whose every ingredient (ge-
3 ometry, per-line gains, noise law) is measured on the instrument; we treat the
4 simulator’s imperfection as the object of study. An ensemble trained across sim-
5 ulators drawn within the measured calibration uncertainty separates simulator
6 uncertainty from initialisation variance ($2.8\times$ wider than a fixed-simulator control
7 on the real scan, 86 % of its variance attributable to the simulator), carries a predic-
8 tive band far closer to calibrated on the real scan than the control’s (0.61/0.87/0.95
9 vs 0.53/0.80/0.91 at $1/2/3\sigma$; both still below the Gaussian 0.68/0.95/0.997), and its
10 mean is the first learned restoration in our study that does not lose correlation to
11 the physics inverse on the real scan. Eighteen deliberately broken training simula-
12 tors show the learned refinement tolerates large noise and gain errors but fails on
13 geometry: 0.5 px of registration error already hurts, 1 px makes the network worse
14 than the physics it refines. A blind diagnostic battery thresholded on the calibration
15 uncertainty identifies every defect that causes material damage and, on the real scan,
16 rediscovers unsupervised the resampling-blur mismatch that originally broke our
17 training pipeline. Used as ABC summaries, the same statistics turn one real scan
18 into a posterior over the simulator: the blur defect is rejected (prior 0.5, posterior
19 0.000) and a posterior-trained ensemble is the study’s most accurate restoration,
20 with a 27 % narrower band than the prior’s. A 180-case degradation grid with four
21 classical inpainting controls shows where learning pays: everywhere on measured
22 pixels, nowhere inside missing regions, where a biharmonic fill wins in 16 of 16
23 cells. All results are anchored on real measurements from one instrument and one
24 painting; all negative results are reported.

25 1 Introduction

26 Machine learning for experimental science usually trains on simulations because ground truth is
27 scarce, and the simulator is never exactly the instrument; the common response is to validate it once
28 and trust it. This paper takes the opposite view on a small but fully measured problem: we ask *how*
29 imperfect a validated simulator still is, what that costs a network trained on it, whether it can be
30 detected and attributed blind, how to represent it in predictive uncertainty, and whether one real
31 measurement can reduce it.

32 Our testbed is the restoration of geometrically degraded MA-XRF element maps of an oil painting
33 [1]: a scan acquired at a tilt must be registered, gain-corrected and denoised back into the frontal
34 frame. The forward model is not designed but measured (Section 2), which gives us something rare:
35 a per-component calibration uncertainty for every knob of the simulator. Our contributions:

- **A documented failure mode of simulator validation.** A forward model can pass a fidelity gate as an emulator and still be wrong as a training-data generator; we document the mechanism (a single resampling blur, Section 2) and reuse it as a controlled defect, diagnostic target and ABC hypothesis.
- **Uncertainty that knows the simulator is imperfect.** Deep ensembles [8] trained across simulator draws within the measured calibration uncertainty, against a fixed-simulator control that isolates initialisation variance (Section 3): domain randomization [12] with measured ranges.
- **A defect-tolerance ladder and a blind audit.** One network per broken simulator quantifies which defects hurt; a summary-statistic battery thresholded on the calibration uncertainty identifies which component is broken without ground truth (Section 4); every defect with material damage is visible to the audit.
- **Closing the loop, as a checked procedure.** The audit statistics are ABC summaries [2, 4] driving an iterative audit-refine loop with an explicit stopping rule; simulation-based calibration (SBC) [10] on synthetic ground truth confirms the mechanism, not only the one real scan (Section 5).
- **A regime map against classical baselines.** A 180-case grid over tilt, missing data and dose, with four classical inpainting controls, locates where learning pays (Section 6); inside missing regions the network loses to a classical fill.

2 Setup: a measured forward model and its failure mode

Instrument and data. Two silicon drift detectors (70 mm² mounted horizontally and 25 mm² at 40° from horizontal, 12.5 μm Be windows) and a tube at 37 kV / 40 μA scan an oil painting; per-pixel spectra are fitted into net-count maps of 8 element lines (Ca, Ti, Fe, Cu, four Pb L), summed over the detectors. Three scans of the *same painting*: two frontal, F_1 and F_2 (60 × 120 px, seven days apart, same geometry), and one real tilted, T (45 × 80, mounted at 7.7°). F_1 is the only training source; F_2 and T are never seen in training.

Forward model. Every ingredient is measured, not modelled: the warp is the affine registration fitted between T and F_1 ; the per-line tilt gains are the measured level response at the reference mounting, extrapolated linearly in angle; the noise is Poisson-consistent with a per-line constant, $\text{Var}(m) = k_{\text{el}}m$ with $k_{\text{el}} = 4.2\text{--}9.0$ calibrated from the $F_1\text{--}F_2$ difference. A fidelity gate holds on all 8 lines: $r(\text{sim}, T) = 0.89\text{--}0.98$, within 0.02 of the $F_1\text{--}F_2$ repeatability ceiling, and levels within 0.7 %. A residual U-Net [9] (469k parameters, zero-initialised head, so the untrained network *is* the physics inverse) refines the deterministic inversion; training pairs are simulated from F_1 at random angles, doses and dropout blocks (Appendix A).

The blur lesson. Our first training simulator emulated the tilted acquisition by bilinearly warping the frontal map. It passed the fidelity gate, yet the network trained on it *over-sharpened* the real scan (contrast ratio up to 1.05, correlation below the physics inverse on 8/8 lines). A real tilted scan is a direct measurement with no resampling blur, so the simulated restoration inputs carried two interpolation blurs where reality has one. Sampling the source with a cubic spline at the exact warp positions (v2) fixed the training distribution. The transferable lesson: *validated as an emulator is not validated as a training-data generator*, and the failure was invisible on simulated tests.

Perturbable simulator. The simulator’s components are exposed as knobs θ : noise-constant scale, tilt-gain slope, per-line gain offsets, mounting angle, registration shift and rotation, interpolation order (cubic vs v1 bilinear), and, in Section 5, per-line noise scales, each with a measured 1σ calibration uncertainty (noise log-sd 0.33 from the session transfer of k ; gain slope 20 %, the mounting angle an upper bound; registration 0.1 px, 0.3° from per-element fit spread). Perturbations apply only to the training simulator; the test-time inverse stays nominal.

3 Uncertainty that knows the simulator is imperfect

We train two ensembles of $N=12$ members each: a *jitter* ensemble, member i trained on a simulator drawn from the calibration uncertainty, and a *control* ensemble with the nominal simulator and the

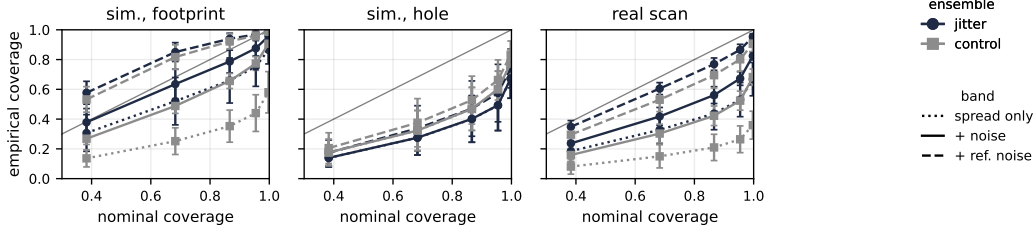


Figure 1: Coverage of $|\text{mean} - \text{truth}| \leq z\sigma$ vs the Gaussian target, jitter (navy) and control (grey); dotted: spread only, solid: plus propagated noise, dashed: plus reference-map noise (sim upper bound; the correct band on the real scan). Bars: sd over lines and cases. Left: 32 simulated cases (footprint); middle: in a 14×20 hole; right: the real scan.

86 same twelve seeds, the standard deep-ensemble baseline [8]. By construction the excess spread of
 87 the jitter ensemble is the variance attributable to simulator uncertainty. The predictive band adds
 88 measurement noise propagated through 8 re-simulations per case plus, on the real scan, the calibrated
 89 noise of the reference map ($k_{\text{el}} \cdot \text{truth}$), since the reference is itself a measurement.

90 **The ensemble knows where the simulator is uncertain.** On the footprint the jitter ensemble spreads
 91 $2.2\times$ (simulated) and $2.8\times$ (real scan) wider than the control; 62 % and 86 % of its variance is
 92 attributable to the simulator draw. The simulator-attributable spread (rms over footprint pixels) grows
 93 monotonically with tilt, from 104 counts at 8° to 148 at 25° , the direction of the gain extrapolation
 94 from the single 7.7° calibration; spatially it follows edges and levels while the control spread lives
 95 only in dropout holes (Appendix Fig. 6).

96 **Calibration.** With spread alone both ensembles under-cover (real scan, $z=2$: 0.53 jitter, 0.26 control).
 97 With the full band the jitter ensemble covers 0.61/0.87/0.95 at $z = 1/2/3$ on the real scan against
 98 0.53/0.80/0.91 for the control (Fig. 1); on simulated cases it is bracketed between 0.64 and 0.85 at
 99 $z=1$ because the reference maps are themselves noisy. This is still under-coverage with heavy tails
 100 (3σ failure rate 0.047 vs nominal 0.003), not calibration; our claim is the difference to the control,
 101 the simulator's share. The jitter spread also ranks the real-scan errors better (Spearman 0.31 vs 0.26,
 102 AUSE [6] 0.33 vs 0.38; binned reliability in Appendix Fig. 7).

103 **Where it fails.** Inside a 14×20 dropout hole neither ensemble is calibrated ($z=2$ coverage 0.49/0.60
 104 against rms error 940 counts vs spread 320–410), the spread does not rank the error (Spearman $\approx$
 105 0.1), and the jitter ensemble is *narrower* than the control (ratio 0.85): all members share the same
 106 image prior and agree on the same wrong fill; simulator uncertainty is not prior uncertainty.

107 **Jitter training costs nothing on the real scan.** The jitter-ensemble mean matches or exceeds the
 108 physics inverse in r on 5/8 lines (mean 0.9527 vs 0.9526, others within 0.002) while restoring contrast
 109 (cv 0.979 vs 0.947); the nominal network and control mean lose r on 8/8 (0.9485, 0.9496). This is a
 110 physically grounded augmentation against the scan's own mismatch, at a price of 0.003 in r where
 111 the nominal simulator is true.

112 4 Auditing the simulator: tolerance and blind diagnostics

113 **Tolerance ladder.** One network per broken simulator, 18 rungs across six families, one shared
 114 seed so that rung differences are the simulator; the 12 control members give the seed band ($\Delta r \in$
 115 $[+0.082, +0.090]$ simulated footprint, $[-0.005, -0.002]$ real). The result (Fig. 2) is asymmetric:
 116 noise can be wrong by $4\times$ either way, the gain slope doubled or removed, the angle belief off by 5° ,
 117 all inside the band. Geometry differs: a 0.5 px shift, five times the calibration uncertainty yet half
 118 a pixel, already leaves the band (four seeds: -0.013 to -0.005 real); at 1 px the network is worse
 119 than the physics inverse it refines (-0.057 simulated, -0.086 real), at 2 px it destroys the restoration
 120 (-0.27 real); a 2° rotation costs -0.047 . The blur rung replays Section 2: invisible on simulated
 121 tests, it leaves the band on the real scan for all four seeds, with the table's only contrast overshoot
 122 (1.013).

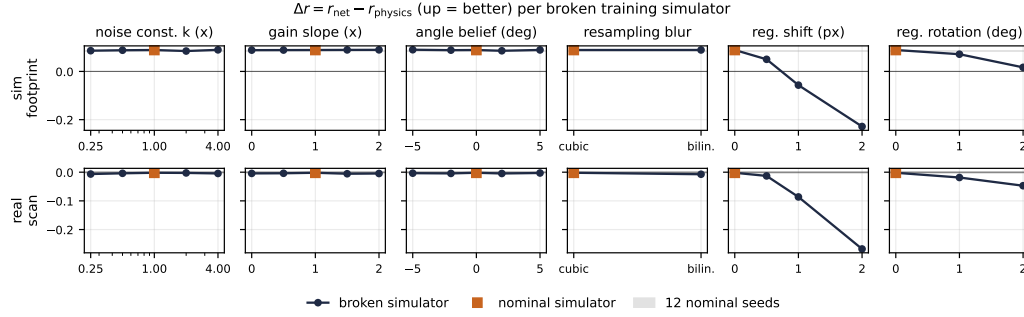


Figure 2: Defect tolerance: $r(\text{net}) - r(\text{physics})$ (up = better), one network per broken training simulator, on nominal simulated cases (top) and the real scan (bottom); grey band: 12 nominal seeds; orange: nominal. The hole region is omitted (seed band $[-0.03, +0.27]$, no tolerance signal). Geometry is the cliff; blur is visible only on the real scan.

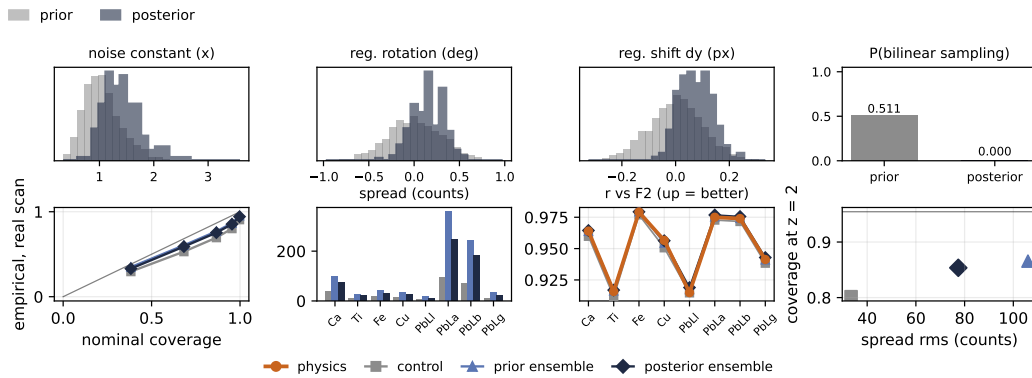


Figure 3: Top: prior (grey) and posterior (navy) ABC marginals from one real scan, for the knobs the data moves (gain slope and angle match the prior; omitted); $P(\text{bilinear})$ drops from 0.5 to 0.000. Bottom: real-scan calibration, per-line spread and r (physics in orange), and the coverage-vs-width tradeoff.

Blind diagnostics. Can the broken component be identified from one scan, without truth? We compare measurement to simulation with nine per-line statistics (level, contrast, mid- and high-band power ratios, a noise-constant estimate, signed sub-pixel offsets, a rotation estimate, a gain-template projection). The null is the crucial choice: with noise-only pairs everything fires on the real scan (a 0.3° registration residual gives $z \approx 60$), so we build it from pairs drawn *within the calibration uncertainty*: a flag means the discrepancy exceeds what the simulator claims to know. With a rule fixed before the experiment, the battery identifies blur 10/10, shifts down to 0.5 px 30/30, rotations of $1-2^\circ$ 20/20 and noise scalings $\times 0.25/2/4$ 30/30, with 4/48 false alarms; gain-family defects within the calibration uncertainty are invisible by construction (0/80; a post-hoc template recovers only the extremes) because they are an exact rank-one degeneracy with the angle belief at one tilt (Appendix B.1). Crossing audit and ladder: *every rung with material damage is visible, every invisible one harmless* (one exception at 0.0002, the band's resolution). On the real scan the battery, blind, flags our v1 simulator and the validated bilinear emulator as “blur” (high-frequency power ratio 1.7–2.1) and leaves the v2 simulator with one finding, a residual noise excess (k ratio 1.19): it rediscovers the blur lesson (Appendix Fig. 8).

5 Closing the loop: a posterior over the simulator from one scan

The audit statistics support likelihood-free inference [4]: rejection ABC [2] over the simulator's knobs, prior = the calibration jitter of Section 3 plus the v1 bilinear sampling at probability 0.5, so the posterior must reject the v1 defect on its own; 3000 draws, whitened RMS- z distance, closest 5 % accepted (2 % and 10 % agree in every direction).

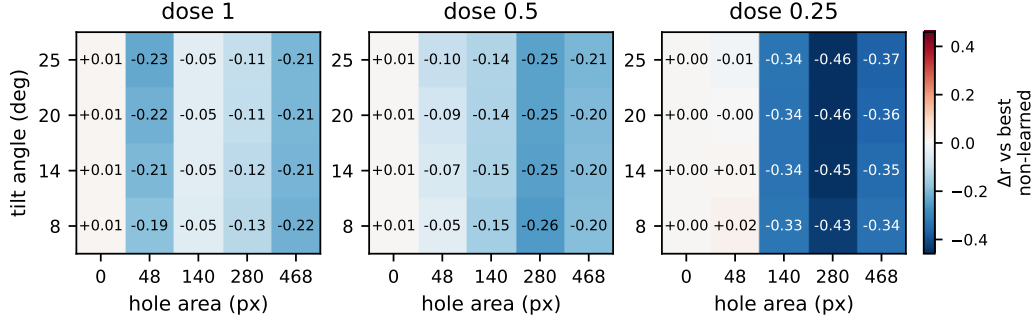


Figure 4: Regime map: $r(\text{net})$ minus the best non-learned candidate (physics or best classical fill) inside the hole (footprint for the no-hole column), per dose; blue = the non-learned candidate wins, red = the network wins. Per-candidate curves in Appendix Fig. 9.

Posterior. The bilinear defect is rejected (posterior 0.000); the posterior concentrates on a noisier instrument (noise scale 1.43 ± 0.35) with a small registration residual ($+0.15^\circ \pm 0.17$, $+0.08 \pm 0.06$ px), gains at calibration. SBC [10] on 24 synthetic trials finds no rank bias in the six continuous knobs and a clean 20/20 discrete blur separation (Appendix B.2): not an artifact of this one scan. On the real scan the predictive check rejects every draw (0/12) with a uniform “noise” verdict from a *per-line* residual (variance ratios 0.9–4.6) a global noise knob cannot express. Formalised as an audit-refine procedure (Appendix B.3), a per-line noise knob narrows the gap (2.16 to 2.02) but an added flat-field knob (documented as unmodelled) does not (2.02 to 1.98, posterior at prior): the loop is judged converged, the residual structural.

Posterior ensemble. Twelve networks retrained on accepted draws are the study’s most accurate real-scan restoration: $r = 0.9539$ against 0.9527 (prior), 0.9526 (physics), 0.9496 (control), at or above the physics inverse on 7/8 lines, with the smallest bias (1.07 %) and a 27 % narrower band (77 vs 106 counts) at coverage 0.85/0.94 at $z = 2/3$ (prior 0.87/0.95). By per-pixel Gaussian NLL, a proper score [5], the wider prior band is marginally better (6.45 vs 6.58 nats; control 7.27), the gap sitting on the lines the knob family cannot express: the score prices what the loop could not fix.

6 When does learning pay? A regime map with classical controls

We sweep 180 cases ($4 \text{ angles} \times 5 \text{ holes} \times 3 \text{ doses} \times 3 \text{ seeds}$, from F_2) and compare physics, the nominal network, the jitter-ensemble mean, and four classical fills (nearest, biharmonic [13], Telea [11], Navier–Stokes [3]), all through the same physics inverse. On measured pixels the learned candidates beat every alternative in every cell (no hole: 0.985 vs 0.977 physics; 14×20 hole footprint: 0.952 ensemble vs 0.937 best classical vs 0.781 physics), with a margin that grows with hole size. Inside the hole the ranking inverts: the biharmonic fill beats the network in 16/16 cells (0.467 vs 0.415 at $20^\circ/14 \times 20$), and at quarter dose the learned fills collapse (0.03 and -0.01) while the classical fill is dose-proof (0.455). The only learned candidate that wins every hole cell is the hybrid, classical fill then network, by 0.005–0.03: the network adds sharpening, not a better fill. A parameter-free spread-weighted blend instead beats the plain ensemble in every hole cell but not this tuned hybrid: the spread under-reacts at low dose (Appendix B.4). Hole columns are angle-independent, since the measured warp does not scale with angle. On this instrument the learned prior buys contrast, noise, neighbourhood repair and calibrated uncertainty, not missing content (Fig. 4).

Limitations and outlook. All of this is one painting, one instrument, one real tilted scan; we claim the auditable chain, not universal numbers, though a synthetic, XRF-unrelated replication reproduces the recipe’s pattern (Appendix B.5), suggesting the benefit is structural. The calibration sigmas are lower bounds (the gain-slope uncertainty is one-sided, symmetrised here); ensembles have 12 members, runs one seed (borderline runs four); 22-dimensional rejection ABC is at its resolution edge (SMC is the upgrade). The hole failure suggests training on classically filled holes; uncertainty-guided acquisition (Appendix Fig. 10) routes to less irradiation of heritage objects, our main foreseen benefit; code and tables will be released.

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A Training and evaluation details

Data and protocol. Frontal maps are 60×120 , the tilted frame 45×80 ; the tilted footprint covers 3466 of the 7200 frontal pixels. Training pairs are simulated from F_1 only: random tilt angle in $[4^\circ, 25^\circ]$, random flips, 25 % low-dose samples ($\times 0.35$ – 1.0), 30 % of samples with 1–2 zeroed dropout blocks of 8–17 px in the tilted frame, recorded in a validity channel. The network input is the deterministic physics inverse of the simulated tilted scan (8 element channels, per-line normalisation fixed from F_1), plus validity and angle channels; the target is the frontal map. Four fixed 15×30 spatial blocks of the frontal frame are excluded from the training loss and serve early stopping. F_2 and the real scan T are never used in training; all final numbers are computed against F_2 as reference, on the footprint and, where applicable, inside the dropout hole. Metrics: Pearson r , SSIM, level bias in percent, and the contrast ratio $CV(\text{candidate})/CV(\text{reference})$.

Architecture and optimisation. Residual U-Net, three resolution levels, base width 32, Group-Norm and SiLU, 469k parameters, 1×1 zero-initialised head (the untrained model equals the physics inverse). Adam [7] with learning rate 10^{-3} , batch 8, at most 1500 steps, validation every 50 steps on a fixed simulated set (8 angles $\times$ 3 replicates), early stopping after 10 validations without improvement, plateau learning-rate decay. One training run takes about 10 minutes on a 12-core CPU workstation; no GPUs were used anywhere. The study trains 60 networks in total (12 jitter, 12 control, 18 ladder rungs, 6 extra-seed variants, 12 posterior members), about 10 CPU-hours, plus inference-only sweeps.

226 Inference itself is cheap: the audit battery and the full 3000-draw rejection-ABC round of Section 5,
 227 from the raw real scan to an updated simulator posterior, run in under three minutes on the same CPU
 228 workstation, with no training and no GPU involved.

229 **Seeds and error bars.** The two ensembles share their twelve seeds, so jitter-vs-control differences
 230 are attributable to the simulator draw; every ladder rung uses the seed of one control member, and the
 231 12 control members define the initialisation band shown in Fig. 2; the two borderline rungs (0.5 px
 232 shift, blur) were re-trained with three extra seeds each and their real-scan damage holds for every
 233 seed. Coverage error bars are standard deviations over lines and cases. The aleatoric band term uses
 234 8 noise re-simulations per case; the reference-noise term uses the calibrated k_{el} .

235 B Further validation and generalization checks

236 B.1 Why gain and angle are degenerate at one tilt

Writing the believed angle as $a = a_0 + \delta_a$ and gain_scale = $1 + \delta_s$, the per-line gain $g_{\text{el}} = 1 + (1 + \delta_s) \frac{p_{\text{el}}}{100} \frac{a}{7.7}$ (with p_{el} the measured tilt response of line el) linearises, at the fixed angle a_0 of one scan, to

$$g_{\text{el}} - g_{\text{el}}^{\text{nom}} \approx (g_{\text{el}}^{\text{nom}} - 1) \left(\delta_s + \frac{\delta_a}{a_0} \right),$$

237 a product of a fixed, measured per-line profile and a single scalar $\kappa = \delta_s + \delta_a/a_0$. At fixed a_0 the
 238 reachable gain vectors form a one-dimensional subspace regardless of how δ_s, δ_a combine, so no
 239 statistic built from one scan can separate the two knobs, only estimate κ (a second scan at a different
 240 angle would: the 2×2 system has determinant $\propto 1/a_0 - 1/a_1 \neq 0$). This is why Section 4's rule
 241 scores gain and angle-belief as one family, why the post-hoc gain-template statistic (the projection
 242 onto the measured profile, the maximum-likelihood estimator of κ) recovers only the rungs with large
 243 $|\kappa|$, and why the ABC posterior of Fig. 3 leaves gain_scale and angle_bias at the prior: one scan at
 244 7.7° cannot see the orthogonal direction.

245 B.2 Simulation-based calibration of the ABC posterior

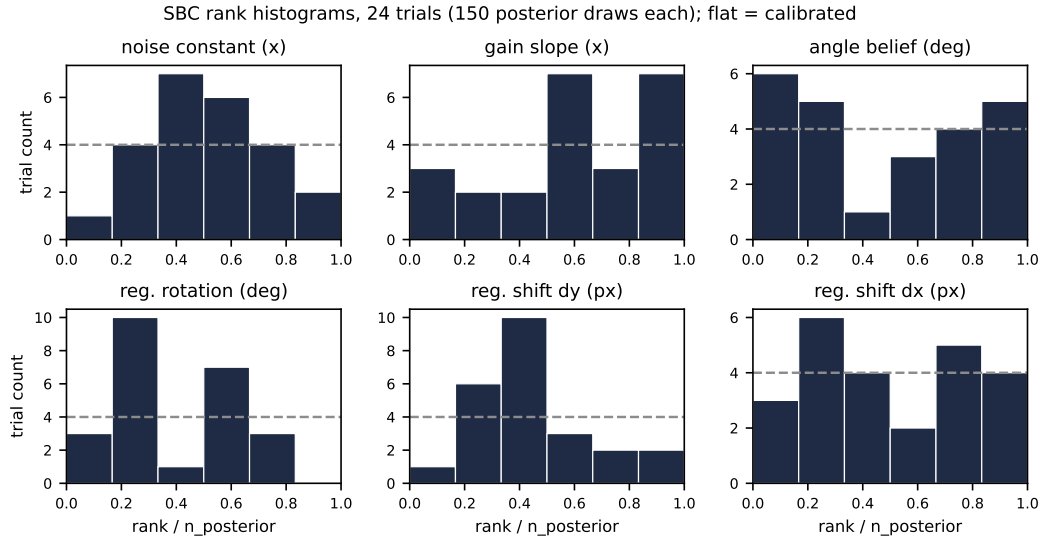


Figure 5: Simulation-based calibration [10]: rank of the true knob value inside 150 posterior draws, over 24 trials with known ground truth (flat = calibrated).

246 Section 5's posterior is plausible on the one real scan, whose true simulator is unknown; that alone
 247 cannot show the loop recovers truth rather than a good-looking story. We draw a true knob vector
 248 from the calibration prior, synthesise a "real" measurement from F_2 under it, run the identical
 249 rejection-ABC procedure, and record the rank of the true value inside the resulting posterior sample.

Over 24 trials (1500 draws each, 10 % accepted) a Kolmogorov-Smirnov test against uniform ranks rejects calibration for 0 of 6 continuous knobs (mean rank 0.41–0.61, all $p > 0.09$, Fig. 5). A parallel discrete test (10 true-bilinear, 10 true-cubic trials, other knobs at nominal) finds posterior $P(\text{bilinear}) = 1.000$ when the truth is bilinear and 0.000 when it is cubic, in all 20 trials. The real scan’s headline finding is therefore not an artifact of that one measurement.

B.3 The audit-refine loop as a procedure

The closed loop is a repeatable procedure with an explicit stopping rule, not a two-shot narrative: a round is added only if motivated by a specific pattern in the previous round’s predictive-check decomposition and corresponds to a documented, physically named mechanism; the loop is judged converged when a round no longer moves the whitened distance beyond null-floor noise or changes the verdict distribution.

round	knob added	$d_{\min}$	null floor	PPC ok/12
0 (nominal)	none	2.72 / 8.92 ^a	0.95 ± 0.24	n/a
1	calibration jitter + blur hypothesis	2.16	0.95 ± 0.24	0/12
2	+ per-line noise scale	2.02	0.95 ± 0.24	0/12
3	+ flat-field (radial)	1.98	0.95 ± 0.24	0/12

^acubic / bilinear simulator. Round 3, motivated by the one remaining unmodelled component the forward model’s own documentation names (a per-pixel flat-field), moves $d_{\min}$ by only 0.04 (round 1 to 2 moved it by 0.14), and its posterior (-0.022 ± 0.053) is statistically indistinguishable from its prior (0.002 ± 0.050): the knob absorbs no signal. By the stopping rule the loop has converged after round 2; the residual is attributed, by elimination across three physically motivated hypotheses, to structure outside this knob family (scatter background, intra-scan dwell drift).

B.4 Parameter-free adaptive blend

The fixed hybrid of Section 6 (biharmonic fill, then always trust the network) is hand-picked. A parameter-free alternative lets the jitter ensemble’s own per-pixel spread decide how much to trust the network: $w_{\text{net}} = \text{var}_{\text{ref}} / (\text{var}_{\text{ref}} + \sigma_{\text{jitter}}^2)$, with var_{ref} fixed a priori as the squared P99 training-scale of each line (computed once from F_1 , no test-case leakage), blended as $w_{\text{net}} \cdot \text{mean} + (1 - w_{\text{net}}) \cdot \text{biharmonic}$. Across the full grid this beats the plain ensemble mean in every hole cell at dose 1 (16/16, mean r 0.391 vs 0.389) but not the tuned hybrid (0/16, 0.391 vs 0.476; tied on the footprint, 8/20). At quarter dose the blend collapses with the network (hole $r = -0.007$) instead of falling back to the dose-proof classical fill (0.455): the ensemble’s simulator-uncertainty spread does not grow enough at low dose to flag the network’s own degradation, a second, independent confirmation that simulator uncertainty is not the same object as fill-trustworthiness uncertainty (Section 3).

B.5 A synthetic, instrument-agnostic sanity check

To check the recipe is not specific to X-ray fluorescence, we repeat it on an unrelated synthetic problem: 32×32 fields built from random Gaussian blobs, degraded by an arbitrary three-knob forward model (multiplicative gain, sub-pixel warp, Poisson-like noise) with synthetic, author-chosen calibration sigmas (stated in the released code, not measured from any instrument). On held-out fields the jitter ensemble’s spread is $8\times$ the control’s, with coverage far closer to nominal at $z = 1, 2$ (0.535/0.826 vs 0.194/0.328 against Gaussian targets 0.683/0.954); a three-statistic blind battery correctly names 18/18 injected out-of-calibration defects; rejection ABC recovers known knob values with a simulation-based-calibration rank close to uniform (mean 0.513 over 60 trial-dimension pairs), with one knob (gain_scale, mean rank 0.660) showing a mild, honestly reported bias. The pattern reproduces on a problem sharing no domain content with the real one, suggesting the recipe’s benefits come from its structure rather than from the X-ray instrument or the painting.

C Additional figures

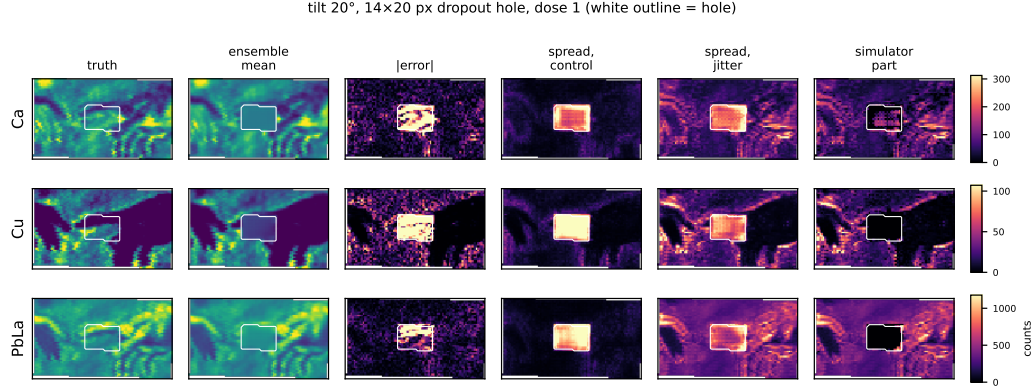


Figure 6: Spread maps on the harsh case (20°, 14 × 20 hole, white outline), three lines. Columns: reference, jitter-ensemble mean, absolute error, control spread, jitter spread, and the simulator part $\sqrt{\max(\sigma_{\text{jitter}}^2 - \sigma_{\text{control}}^2, 0)}$; the three spread panels of a line share one colour scale. The control spread lives in the hole; the jitter spread follows the painting's structure; inside the hole the jitter ensemble is narrower than the control (dark).

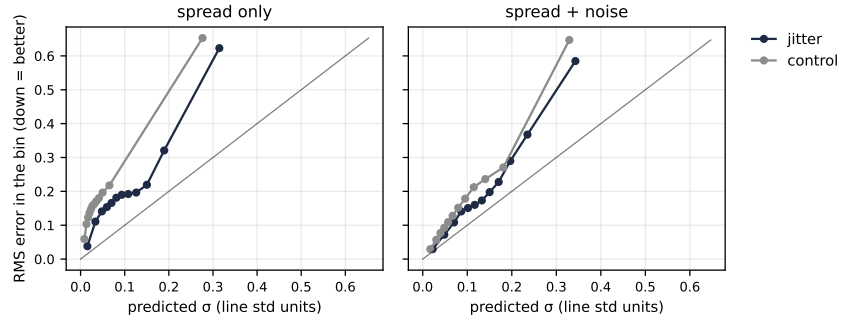


Figure 7: Binned reliability: RMS error against predicted σ in twelve σ -quantile bins, pooled over simulated cases and lines in units of each line's standard deviation; diagonal = calibrated. Left: ensemble spread only; right: spread plus propagated noise.

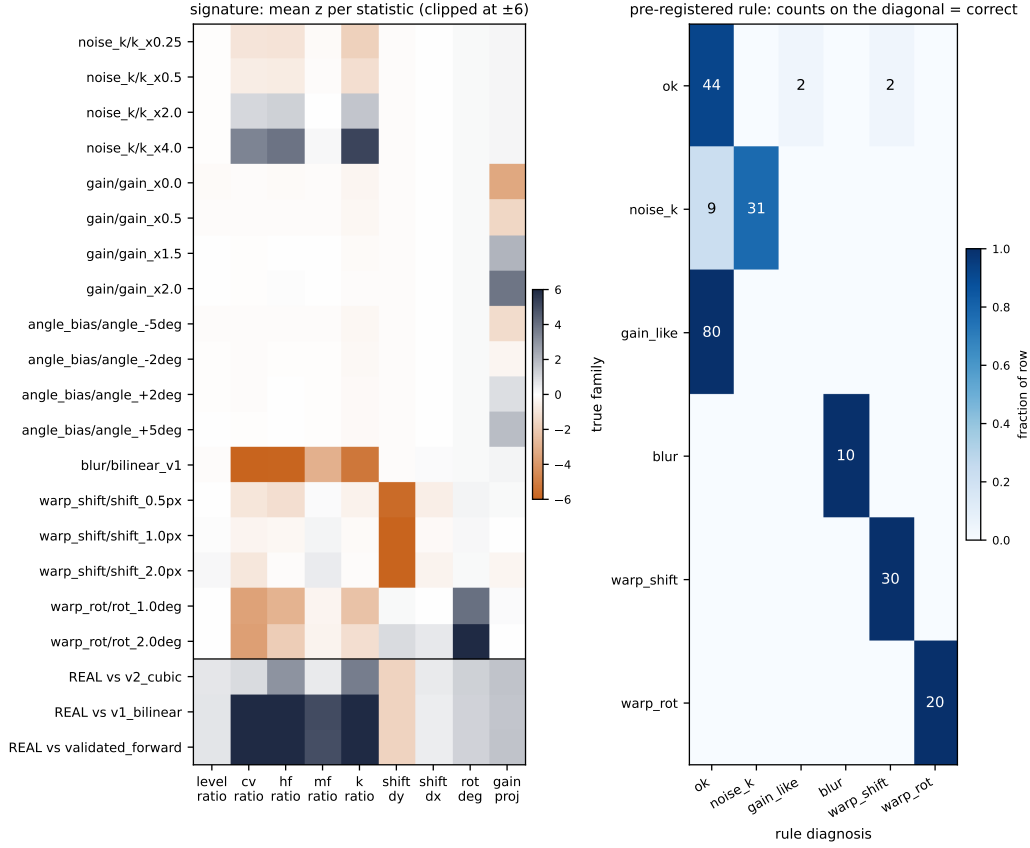


Figure 8: Blind diagnostics. Left: signature of every defect rung, the mean z of each statistic against the calibration-uncertainty null (both test angles pooled, clipped at ± 6); the bottom block is the real scan against three simulators (v2 cubic, v1 bilinear, validated bilinear emulator). Right: identification counts of the pre-registered rule, true family (grouped) versus diagnosis.

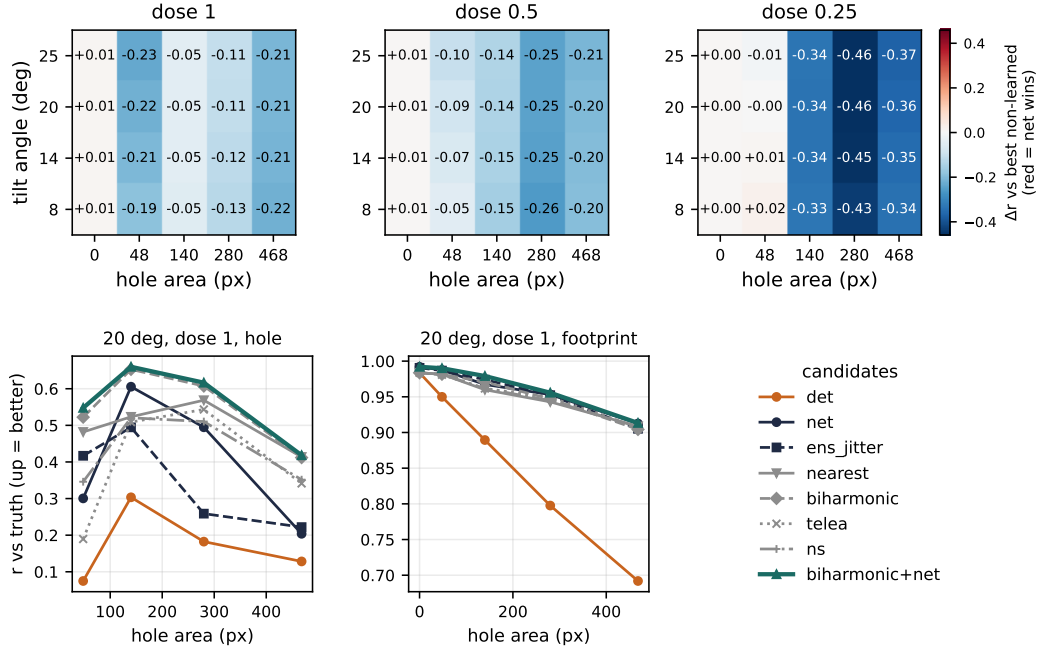


Figure 9: The full regime map: the heat maps of Fig. 4 together with r against hole area at 20° , dose 1, for every candidate, inside the hole and on the footprint (physics orange, learned navy, classical grey, hybrid navy triangles).

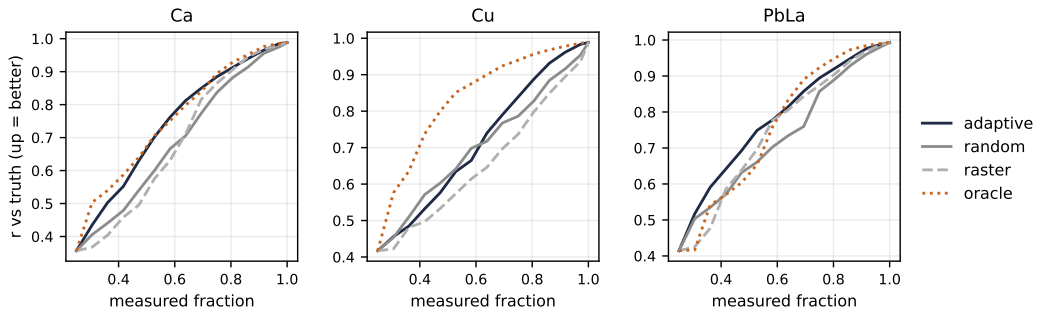


Figure 10: Uncertainty-guided acquisition pilot: r against the measured fraction of the tilted frame when 5×5 tiles are acquired in the order of the jitter-ensemble spread (adaptive), at random, in raster order, or by the true error (oracle, not realisable); 8 tiles per step from a 25 % start, mean of three simulated cases. Adaptive reaches $r \geq 0.90$ (mean over Ca, Cu, Pb L α) after measuring 79 % of the frame, against 85 % for random or raster order and 72 % for the oracle; for $r \geq 0.95$ the fractions are 88 %, 93 % and 82 %.

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